



F20F0 CW2 – DIGITAL FORENSICS COURSEWORK 2 2025-2026



15 APRIL 2026

Shyam Sundar Velmurugan, Harikumar Nandakumar, Joshua Jacob
GROUP - 28
CAMPUS - DUBAI

AUTHORSHIP DECLARATION

Course code and name	F20FO - Digital Forensics
Type of assessment	Group Coursework
Coursework title	F20FO Coursework 2
Student name	Harikumar Nandakumar Shyam Sundar Velmurugan Joshua Jacob
Student ID	H00448161 H00418621 H00404266

By signing this form:

I declare that the work I have submitted for individual assessment or the work I have contributed to a group assessment is entirely my own. I have not taken the ideas, writings or inventions of another person and used them as if they were my own. My submission or my contribution to a group submission is expressed in my own words. Any uses made within this work of the ideas, writings or inventions of others, or of any existing sources of information (books, journals, websites, etc.) are properly acknowledged and listed in the references and/or acknowledgements sections.

I confirm that I have read, understood and followed the University's Regulations on academic misconduct as published on the [University's website](#), and that I am aware of the penalties that I will face should I not adhere to the University Regulations.

I confirm that I have read, understood and avoided the different types of plagiarism explained in the University guidance on [Academic Integrity and Plagiarism](#).

Student signature: Hari Kumar, Shyam Velmurugan, Joshua Jacob

Date: 15/04/2026

Your work will not be marked if a signed copy of this form is not included with your submission.

Contents

Authorship Declaration	1
OneDrive file link for the output image file	3
Introduction	3
Evidence Description	3
Image Processing Using FTK Imager (For Windows)	4
Image processing using Guymager (For Linux)	5
Forensic Analysis using Autopsy (Windows)	6
Forensics with Foremost (Linux)	7
Integrity verification	8
FTK Imager Vs Guymager	9
Autopsy Vs Foremost	9
Conclusion	9
References	10
Appendix	11

OneDrive file link for the output image file

Link: [F29FO CW2 - Digital Forensics Coursework 2](#)

Introduction

Digital forensics plays a crucial role in cybersecurity by focusing on factors like identification, collection and analysis of digital evidence. With specialized tools and techniques, forensic investigators can recover deleted files, check system logs and track user activities for creating a detailed timeline of events, and it becomes an essential when it comes to investigating cybercrimes, security breaches, or any unauthorized access to systems.

This report presents forensic acquisition and analysis of a digital storage device provided as evidence. The objective of the investigation was to create a forensically sound image of the evidence, preserve its integrity and perform a structured analysis to identify relevant artifacts. Industry-standard forensic tools, including FTK Imager, Guymager, Autopsy, and Foremost were used for analysis, under two VMs, Windows and Linux.

Evidence Description

The initial phase of the investigation involved acquiring VMDK from the zip file of digital evidence, which also contained associated data and virtualized operating system. Both FTK Imager and Guymager was used to generate a forensic image of the evidence in .img format, ensuring bit-by-bit copy to maintain integrity. By this process, checksums (MD5, SHA-1 for FTK Imager and SHA-256 for GuyMager) were calculated both before and after processing to confirm that no alterations had occurred.

The VMDK structure consisted of multiple segmented files, indicating that the virtual disk was stored in a split format. These segments collectively represent the full disk image and must be handled carefully to ensure accurate reconstruction during forensic processing. The analysis was conducted on verified copies of the evidence rather than original source, to ensure forensic soundness.

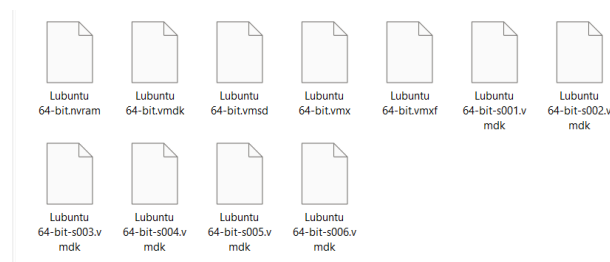


Figure 1 Digital Evidence in Windows

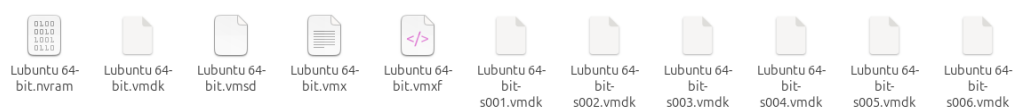


Figure 2 Digital Evidence in Linux

Image Processing Using FTK Imager (For Windows)

In Windows, a forensic image of the evidence was created using FTK Imager. The tool is widely recognised in digital forensics for the ability to generate exact bit-by-bit copies of digital storage media while preserving the integrity of the original data (*Lim et al. (2025)*). The user-friendly interface and built-in verification mechanisms make it suitable for reliable evidence acquisition in a controlled forensic environment.

The process was performed by selecting “Create Disk Image” option and choosing image file as the source type. The source path of the VMDK evidence was specified, after which the process was initiated by confirming the selection. Adding image destination, where the output format was configured as raw (dd) to ensure complete bit-stream copy of the evidence. Relevant evidence item information, including case details and examiner data, was entered to maintain proper documentation. The destination folder for storing forensic image was selected, and the configuration was finalised before starting the acquisition process. Once image was completed, cryptographic hash values, including MD5 and SHA-1, were verified to confirm the integrity and authenticity of generated image.

During the acquisition process, FTK Imager performed a sector-by-sector duplication of the evidence, ensuring that all data, including active, deleted, and unallocated space, was preserved. The use of cryptographic hashing provided reliable mechanism to validate that the acquired image was identical to the original source and hash values also matched with before and after imaging. Both MD5 and SHA-1 before and after values matching confirms that no alterations had taken place during the process.

The output generated from FTK Imager includes a raw forensic image file along with associated hash values and acquisition details. These outputs form critical part of chain of custody, as they prove that the evidence has been handled in forensic manner. The use of FTK Imager in this investigation established a trusted baseline for further analysis, ensuring that all subsequent forensic procedures were conducted on verified and unaltered copies of original evidence.

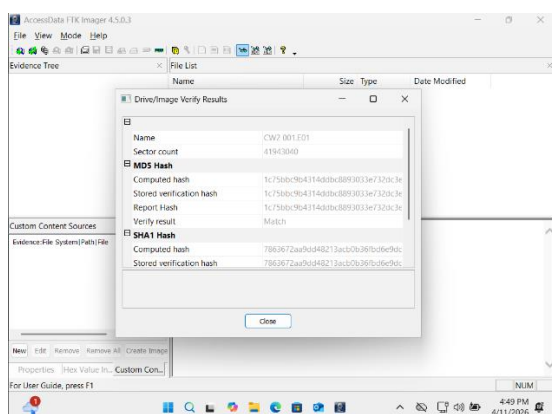


Figure 3 MD5 hashing after image processing

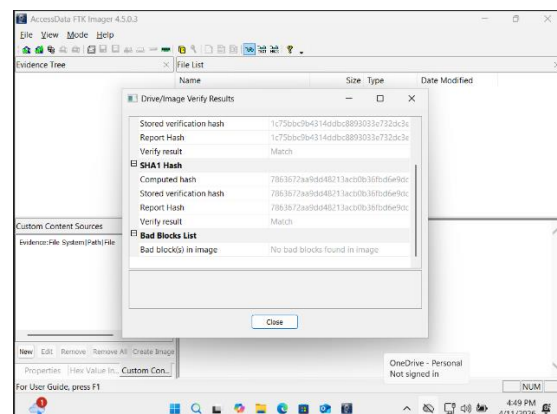


Figure 4 SHA-1 Hashing after image processing

Image processing using Guymager (For Linux)

Forensic imaging on Linux was created using Guymager within a Linux-based environment. This is an open-source imaging tool designed for efficient and reliable acquisition of digital evidence, supporting multiple hashing algorithms and built-in verification mechanisms (Himansu *et al.* (2021)). The VMDK evidence files were made accessible within the forensic virtual machine through a shared folder configuration.

The process starts by selecting the evidence device from the device list, and the acquisition process was configured accordingly. The output format was set to raw (dd) for a complete bit-by-bit copy of the evidence. Split size was explicitly set to 0 so that the output image would not be divided into multiple segments, thereby ensuring that the forensic image remains as a single continuous file, simplifying the analysis and preventing any type of potential integrity or reconstruction issues during handling or transfer. Hashing was enabled using both MD5 and SHA-256 algorithms to support integrity verification, and automatic verification after acquisition was selected to validate the image upon completion. Additionally, the output directory and filename were specified, and relevant case details such as case number, evidence ID, examiner name, and description were entered to maintain proper documentation.

Following configuration, image process was initiated, and Guymager performed a full bit-stream duplication of evidence, capturing all data including allocated, unallocated, and deleted regions. During acquisition hash values were calculated in real time to ensure that the data remained unchanged. Upon completion, Guymager automatically compared hash values of source and acquired image. The matching results confirmed that forensic image was an exact and unaltered replica of original evidence.

The process generated multiple output files, including a .dd raw forensic image file and a corresponding .info metadata file containing acquisition details and hash values.

```
Linux device      : /dev/sdX
Device size      : 53153792 (53.2MB)
Format           : Linux dd raw image - file extension is .dd
Image path and file name : /home/vboxuser/OUTPUTCWE/finaltest.dd
Info path and file name : /home/vboxuser/OUTPUTCWE/finaltest.info
Hash calculation  : MD5 and SHA-256
Source verification : off
Image verification : on

No bad sectors encountered during acquisition.
State: Finished successfully

MD5 hash          : dha65756269be5791f8467d426db596d
MD5 hash verified source : --
MD5 hash verified image  : dha65756269be5791f8467d426db596d
SHA1 hash         : --
SHA1 hash verified source : --
SHA1 hash verified image  : --
SHA256 hash       : 66fa60b041cd05d0b2cd27ba01bfaf1a02a5ff8c7d8b01fb0637666be153
SHA256 hash verified source : --
SHA256 hash verified image  : 66fa60b041cd05d0b2cd27ba01bfaf1a02a5ff8c7d8b01fb0637666be153
Image verification OK. The image contains exactly the data that was written.

Acquisition started : 2026-04-11 23:49:32 (ISO format YYYY-MM-DD HH:MM:SS)
Verification started: 2026-04-11 23:49:35
Ended               : 2026-04-11 23:49:36 (0 hours, 0 minutes and 3 seconds)
Acquisition speed   : 25.35 MB/s (0 hours, 0 minutes and 2 seconds)
Verification speed   : 50.69 MB/s (0 hours, 0 minutes and 1 seconds)

Generated image files and their MD5 hashes
=====
No MD5 hashes available (configuration parameter calc_image_file_md5 is off)
MD5 Image file
n/a finaltest.dd
```

Figure 5 Comparing before and after MD5 and SHA-256 values after image processing

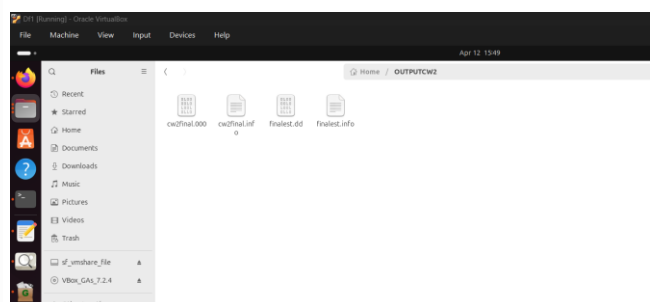


Figure 6 the output .dd file and info file

Forensic Analysis using Autopsy (Windows)

The forensic analysis on Windows was conducted using Autopsy, a digital forensics platform designed to examine disk images and reconstruct file systems for investigative purposes (*The Cool One (2020)*). The analysis process started with creation of a new case, where details like case number and storage location were specified to organise the investigation. The evidence was added using the “Disk Image or VM File” option, and the VMDK file was selected as the data source for analysis.

During ingest configuration stage, only essential modules were enabled to optimise performance and manage storage constraints. These included File Type Identification, Keyword Search, and Recent Activity, while all other non-essential modules were disabled to reduce overhead processing. Ingest process was then initiated, allowing Autopsy to analyse evidence and reconstruct file system structure.

Once processing began, the reconstructed file system was explored through the DATA Sources and File Views section, enabling systematic navigation of directories and files. Deleted files were identified through filesystem analysis and examined to determine their relevance. Individual files were opened and analysed in detail, including inspection of file contents and associated metadata such as timestamps and the file paths, which are critical for understanding the sequence of system activities and user behaviour.

During the analysis certain files were automatically flagged by Autopsy as potential points of interest. These included “bad” items, indicated by a red circle with exclamation mark, and “suspicious” items, indicated by a yellow circle with inverted triangle. These indicators assisted in prioritising files for further examination. Relevant files identified during the investigation were tagged accordingly to organise important evidence and support structured reporting.

The use of Autopsy enabled efficient and structured forensic analysis by combining automated processing with manual inspection, facilitating identification of key artifacts while maintaining integrity and traceability of evidence.

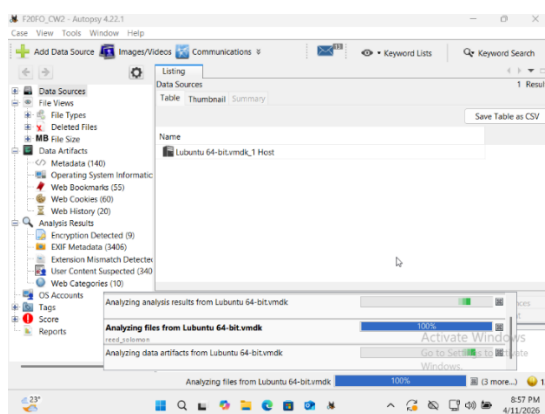


Figure 7 Running ingestion modules for file analysing

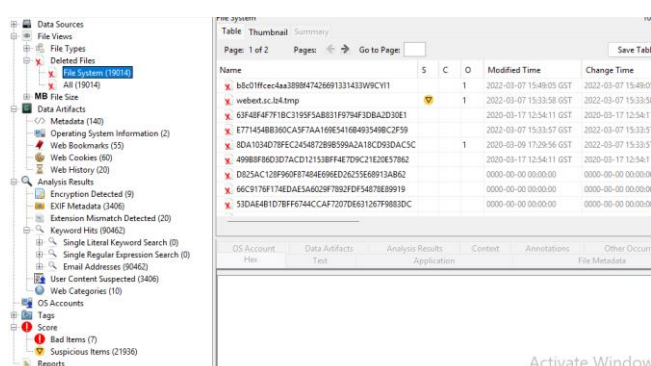


Figure 8 Showing deleted files, suspicious and bad items

Forensics with Foremost (Linux)

Foremost is a file carving forensic tool used to recover deleted or hidden files directly from raw disk images. It operates by scanning disk data for known file signatures, such as headers and footers, and extracting recoverable files without relying on the existing file system structure. This makes it particularly effective in situations where file system metadata is missing, corrupted, or insufficient for recovery. Foremost is widely used in digital forensics for recovering deleted evidence, extracting files from damaged images, and identifying hidden artifacts that may not be detected by conventional analysis tools.

In this investigation, Foremost was utilised to perform deep-level recovery of deleted or residual files from the forensic image generated during the acquisition phase (*Kornblum (2006)*). Following the creation of the forensic image *finalest.dd* using Guymager, Foremost was executed through the command-line interface to analyse raw image data.

The process involved configuring Foremost to scan all supported file types using the `-t all` parameters. The input forensic image was specified using `-i` option, while the output directory was defined using the `-o` option. Additionally, verbose mode `-v` was enabled to display detailed processing information during the execution. This configuration allowed Foremost to perform a comprehensive scan of the entire 50 MB forensic image.

Upon completion, Foremost generates output directory containing structured folders which consists of EXE folder storing all recovered executable files, PNG folder containing all recovered image files, and an *audit.txt* file that documents entire recovery process, including offsets, sizes, and timestamps. Access to these recovered files is restricted, requiring appropriate permissions. This also shows that Foremost was able to retrieve files which may not be visible to traditional filesystem-based analysis tools such as Autopsy, thereby highlighting its effectiveness in identifying residual data directly from disk-level structures.

```
vboxuser@pfi:~$ sudo foremost -t all -i /home/vboxuser/OUTPUTQ2/finaest.dd -o outputQ2 -v
Foremost version 1.5.7 by Jesse Kornblum, Kris Kendall, and Nick Mikus
Audit File

Foremost started at Sat Apr 11 23:58:27 2026
Invocation: foremost -t all -i /home/vboxuser/OUTPUTQ2/finaest.dd -o outputQ2 -v
Output directory: /home/vboxuser/outputQ2
Configuration file: /etc/foremost.conf
Processing: /home/vboxuser/OUTPUTQ2/finaest.dd
-----
File: /home/vboxuser/OUTPUTQ2/finaest.dd
Start: Sat Apr 11 23:58:27 2026
Length: 50 MB (53153792 bytes)

Num   Name (bs=512)      Size   File Offset Comment
-----
0:    00000160.exe      241 KB      86016 12/17/2024 10:24:11
1:    00013376.exe      1815 KB     6848512 10/17/2025 09:55:45
2:    00015400.exe        611 KB     7025760 10/17/2025 09:55:34
3:    00016772.exe        218 KB     8587264 10/17/2025 09:55:34
4:    00017276.exe        181 KB     8845312 10/17/2025 09:55:34
5:    00017084.exe        827 KB     9054208 10/17/2025 09:55:34
6:    00012494.exe        563 KB     9334840 10/17/2025 09:55:34
7:    00028060.exe        214 KB     10551296 10/17/2025 09:55:34
8:    00021084.exe        241 KB     10795808 12/17/2024 10:24:11
9:    00038060.exe        241 KB     19498816 12/17/2024 10:24:11
10:   00000610.png         8 KB       312744 (256 x 256)
11:   00014975.png         8 KB       7667424 (256 x 256)
12:   00016293.png         8 KB       8342260 (256 x 256)
13:   00018987.png         8 KB       9688628 (256 x 256)
14:   00021520.png         8 KB      11021736 (256 x 256)
15:   00038510.png         8 KB      19717544 (256 x 256)
*]
Finish: Sat Apr 11 23:58:28 2026

16 FILES EXTRACTED

exe:= 10
png:= 6
-----
Foremost finished at Sat Apr 11 23:58:28 2026
```

Figure 9 Command to generate output directory containing all structured folders for recovered files

Integrity verification

Integrity verification was performed during both acquisition processes using cryptographic hash functions. FTK Imager generated MD5 and SHA1 hashes, while Guymager generated MD5 and SHA-256 hashes. These hash values serve as unique digital fingerprints of the data.

Matching hash values between the source and the acquired images confirmed that no alterations occurred during the imaging process. This verification step is critical in ensuring that the forensic image remains an exact representation of the original evidence.

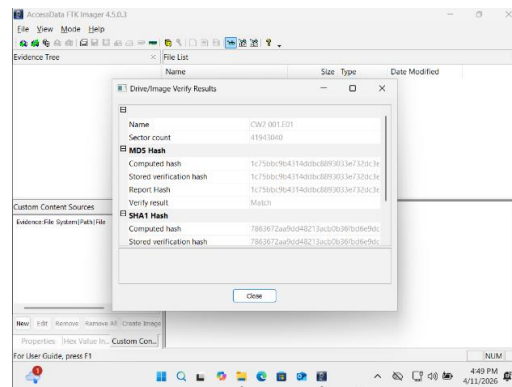


Figure 10 MD5 and SHA-1 hash values from FTK-Imager

```
MD5 hash : 4ba65756269be5791f8467d426db596d
MD5 hash verified source : --
MD5 hash verified image : 4ba65756269be5791f8467d426db596d
SHA1 hash : --
SHA1 hash verified source : --
SHA1 hash verified image : --
SHA256 hash : 66fa60b041fcd5d8b2ed22ba91bfafafaa3a5ff05c7d8ba01fbbe639669e153
SHA256 hash verified source: --
SHA256 hash verified image : 66fa60b041fcd5d8b2ed22ba91bfafafaa3a5ff05c7d8ba01fbbe639669e153
Image verification OK. The image contains exactly the data that was written.
```

Figure 11 MD5 and SHA-256 hash values from Guymager

The results provided by Foremost not only proves that it had performed file carving successfully but also proved that recovering artifacts that may not be visible through the traditional filesystem-based analysis tools such as Autopsy.

In the context of the investigation, Foremost complemented the analysis performed using Autopsy by providing an additional layer of evidence recovery. It enabled the identification of potentially deleted or unindexed files, thereby enhancing the completeness and reliability of the forensic investigation. Overall, the use of Foremost validated the presence of recoverable artifacts at the raw data level and strengthened the overall evidential findings.

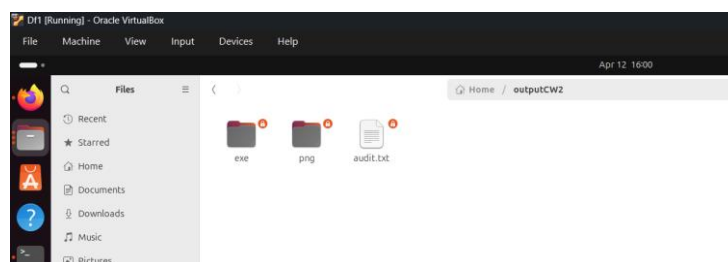


Figure 12 the EXE, PNG, Audit.txt files after running the command in CLI using Foremost

FTK Imager Vs Guymager

Feature	FTK Imager	Guymager
Platform	Windows	Linux
Interface	Graphical (GUI)	Graphical (GUI)
Output Format	Raw (dd), E01	Raw (dd), EWF
Hash Algorithms	MD5, SHA1	MD5, SHA-256
Verification	Built-in	Built-in (automatic)
Metadata Logging	Basic	Detailed
Performance	Moderate	High (optimized for Linux)
Cost	Free	Open source

Autopsy Vs Foremost

Aspect	Autopsy	Foremost
Platform	Windows	Linux
Purpose	Full forensic analysis of disk images	Recovery of deleted/hidden files
Analysis Level	High-level + structured investigation	Low-level raw data recovery
Method	Filesystem-based analysis	Signature-based file carving
Interface	GUI-based tool	CLI-based tool
Deleted Files Handling	Uses metadata and filesystem records	Recovers directly from disk blocks
Output Type	Organized case data and artifacts	Extracted files in category folders
Strength	Investigation + correlation of evidence	Finds hidden/deleted data missed by others
Limitation	May miss deeply deleted data	No contextual analysis (no timeline/metadata)

Conclusion

This investigation demonstrated a structured and forensically sound approach to digital evidence in the form of a VMDK, the VMDK was first preserved using forensic imaging techniques to ensure data integrity, with hashing mechanisms confirming that no modifications occurred during acquisitions. Imaging was performed in both Windows and Linux, ensuring cross-platform validation of the forensic process.

Subsequent analysis using Autopsy enabled systematic examination of the file system, recovering artifacts, and identification of user activity, while Foremost provided deeper file-level recovery by carving deleted or residual data directly from raw disk content. The combination of all four tools ensured a layered forensic methodology, where each tool contributed a distinct investigative strength, ultimately improving evidence completeness and reliability. Overall, this investigation confirms that using multiple forensic tools in a structured workflow significantly enhances the robustness, accuracy and credibility of digital forensic analysis.

References

- [1] The Cool One (2020). Autopsy Tutorial for Digital Forensics. [online] Medium. Available at: <https://medium.com/@tusharcool118/autopsy-tutorial-for-digital-forensics-707ea5d5994d>.
- [2] Himanshu, S. Bhatt and G. Garg, "Comparative analysis of acquisition methods in digital forensics," 2021 Fourth International Conference on Computational Intelligence and Communication Technologies (CCICT), Sonapat, India, 2021, pp. 129-134, doi: 10.1109/CCICT53244.2021.00035.
- [3] Kornblum, J. (2006). Identifying almost identical files using context triggered piecewise hashing. Digital Investigation.
- [4] Lim, Michelle & Chen, Brandon & Lim, Le Ying & Shanbagamaran, Tarini & Lim, Darren & Hlaing, Ts. Dr. Ngu & Sahban Rafsanjani, Ahmad. (2025). Analysis of Forensic Disk Imaging Tools for Data Acquisition and Preservation. Journal of Informatics and Web Engineering. 4. 158-181. 10.33093/jiwe.2025.4.2.11.

Appendix

Chain of Custody

S. No.	Date	Examiner	Action Done	Remarks
1	11/04/2026	Joshua Jacob	Received VMDK evidence from compressed archive	Evidence obtained in original state; integrity assumed intact
2	11/04/2026	Joshua Jacob	Verified evidence integrity using MD5 and SHA-1 hashing	Hash values recorded before imaging; no alteration detected
3	11/04/2026	Joshua Jacob	Created forensic image using FTK Imager	Raw (dd) image generated with verified hash consistency
4	11/04/2026	Harikumar	Created secondary forensic image using Guymager.	Cross-platform imaging performed; MD5 and SHA-256 verified
5	12/04/2026	Team	Transferred forensic image for analysis	Evidence handled in controlled forensic environment
6	12/04/2026	Shyam Sundar	Loaded image into Autopsy for analysis	Ingest modules configured; filesystem reconstruction performed
7	12/04/2026	Shyam Sundar	Choose the ingestion modules and run the ingestions.	Ingestion modules run and file architecture was created.
8	12/04/2026	Shyam Sundar	The file architecture was analysed	Found different file types, deleted files, and score.
9	12/04/2026	Shyam Sundar	Identified deleted files and analysed artifacts	Files examined using metadata and file structure views
10	12/04/2026	Shyam Sundar	Analysed files that had either Bad or Suspicious tag on them.	Each file is examined to understand why those tags are given.
11	12/04/2026	Joshua Jacob	Performed file carving using Foremost	Deleted and residual files successfully recovered
12	12/04/2026	Joshua Jacob	Tagged and documented relevant artifacts	Evidence categorized for reporting
13	12/04/2026	Team	Generated forensic findings report	Final analysis completed with supporting artifacts
14	12/04/2026	Team	Secured and archived evidence	Chain of custody maintained throughout investigation